

Directions. Complete all the problems. Your solutions to the problems should be submitted to Gradescope before the indicated due date above. Please follow rules regarding Gradescope submission as described in the syllabus. If you use any outside source in your solutions, you must cite the source.

Problem 1. d'Alembert's Solution to the Wave Equation

Consider the one-dimensional wave equation for an infinite string:

$$u_{tt} = c^2 u_{xx}, \quad -\infty < x < \infty, \quad t > 0$$

- (a) Verify by direct differentiation (using the chain rule) that $u(x, t) = F(x + ct) + G(x - ct)$ is a general solution to the wave equation, where F and G are arbitrary twice-differentiable functions.
- (b) Now, consider the initial value problem (IVP) for an infinite string with a given initial displacement and zero initial velocity:

$$\begin{cases} u_{tt} = c^2 u_{xx} \\ u(x, 0) = f(x) \quad u_t(x, 0) = 0 \end{cases}$$

Show that the solution to this specific problem is given by d'Alembert's formula:

$$u(x, t) = \frac{1}{2}[f(x + ct) + f(x - ct)]$$

Hint: Use the general solution from Part (a) and solve for F and G using the initial conditions. You may need to integrate $G'(x) = -F'(x)$.

- (c) Use the result from Part (b) to solve for the displacement $u(x, t)$ given the "triangular pulse" initial condition:

$$f(x) = \begin{cases} 1 - |x| & \text{if } |x| \leq 1 \\ 0 & \text{if } |x| > 1 \end{cases}$$

Sketch (or use graphing software) the profile of the string $u(x, t)$ at times $t=0$, $t=1/2c$, and $t=1/c$. Briefly describe what is happening.

Problem 2. The method of separation of variables relies on solving an eigenvalue problem. Consider the spatial part of the heat/wave equation on a finite domain $[0, L]$, which leads to the Sturm-Liouville problem:

$$X''(x) + \lambda X(x) = 0$$

Find the eigenvalues λ and corresponding eigenfunctions $X_n(x)$ for this equation subject to the following boundary conditions:

- (a) **Homogeneous Neumann conditions (insulated ends):** $X'(0) = 0$, $X'(L) = 0$. (These are used for a rod with insulated ends.)
- (b) **Mixed (Dirichlet-Neumann) (fixed at $x = 0$, insulated at $x = L$):** $X(0) = 0$, $X'(L) = 0$
- (c) **Mixed (Neumann-Dirichlet) (insulated at $x = 0$, fixed at $x = L$):** $X'(0) = 0$, $X(L) = 0$

Problem 3. Let $u_t = a^2 u_{xx}$ on $0 < x < L$, $t > 0$ with homogeneous Dirichlet boundary conditions $u(0, t) = u(L, t) = 0$. Let the initial condition be

$$u(x, 0) = f(x) = \begin{cases} x, & 0 \leq x \leq L/2, \\ L - x, & L/2 < x \leq L. \end{cases}$$

- (a) Find the eigenvalues and eigenfunctions.
- (b) Compute the Fourier sine coefficients for $f(x)$.
- (c) Write the solution as a Fourier sine series.

Problem 4. Consider

$$\begin{cases} u_t = u_{xx}, & 0 < x < \pi, \quad t > 0, \\ (u(0, t) = 0, & u(\pi, t) = 3 \\ u(x, 0) = \frac{3x}{\pi} + \sin x \end{cases}$$

- (a) Find the steady state (time-independent) solution $v(x)$ that satisfies the boundary conditions.
- (b) Define $w(x, t) = u(x, t) - v(x)$. Write the PDE and boundary conditions satisfied by w , showing they are homogeneous on the boundary.

- (c) Using separation of variables, write the series solution for w and hence give $u(x, t)$. (You do not need to evaluate every coefficient explicitly — give the formula for coefficients.)

Problem 5. Solve

$$\begin{cases} u_t = a^2 u_{xx}, & 0 < x < L, \quad t > 0, \\ u_x(0, t) = 0 & u_x(L, t) = 0 \\ u(x, 0) = \cos\left(\frac{\pi x}{L}\right). \end{cases}$$

Give the explicit solution and comment on the presence (or absence) of the zero eigenmode.

Problem 6. Consider

$$\begin{cases} u_t = u_{xx}, & 0 < x < 1, \quad t > 0 \\ u_x(0, t) = 0 \\ u_x(1, t) + \alpha u(1, t) = 0, & \alpha > 0, \\ u(x, 0) = 1 - x \end{cases}$$

- (a) Explain briefly how the eigenvalue problem changes compared to Dirichlet/Neumann cases.
- (b) Write the formal eigenfunction expansion for the solution (no need to compute coefficients explicitly).

Problem 7. Solve the wave equation

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < L, \quad t > 0,$$

with $u(0, t) = u(L, t) = 0$ and initial data

$$u(x, 0) = \sin\left(\frac{3\pi x}{L}\right), \quad u_t(x, 0) = 0.$$

Give the solution.

Problem 8. Let

$$u_{tt} = u_{xx}, \quad 0 < x < \pi, \quad t > 0,$$

with $u(0, t) = 0$ and $u_x(\pi, t) = 0$. Suppose

$$u(x, 0) = \begin{cases} 0, & 0 \leq x < \frac{\pi}{2}, \\ 1, & \frac{\pi}{2} \leq x \leq \pi, \end{cases} \quad u_t(x, 0) = 0.$$

Describe how to set up the eigenfunctions (sine/cosine variants) for this mixed problem and write the solution series (coefficients can be left as integrals).

Problem 9. Solve

$$u_{tt} = 4u_{xx}, \quad 0 < x < \pi, \quad t > 0,$$

with $u(0, t) = u(\pi, t) = 0$ and initial conditions $u(x, 0) = 0$, $u_t(x, 0) = 2 \sin(3x)$. Write $u(x, t)$ explicitly.

Problem 10. Solve the heat equation on the rectangle $0 < x < a$, $0 < y < b$:

$$u_t = a_0^2 (u_{xx} + u_{yy}), \quad t > 0,$$

with zero boundary conditions $u = 0$ on the boundary and initial condition

$$u(x, y, 0) = \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi y}{b}\right).$$

Give the solution and state the decay rate in time of this mode.

Problem 11. Let $0 < x < \pi$, $0 < y < \pi$. Solve

$$u_t = u_{xx} + u_{yy},$$

with $u = 0$ on the boundary and initial data

$$u(x, y, 0) = x \sin(y).$$

- (a) Expand the initial data in the double sine series (write coefficients as integrals).
- (b) Write the general series solution and identify which terms will decay slowest.